CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 5: **Describing Motion Around Us** | Quick revision, one page

Distance, displacement, speed, velocity

- Distance: total path length (scalar, SI unit m). Displacement: shortest straight-line distance from initial to final position with direction (vector, m).
- Average speed $= \dfrac{\text{total distance}}{\text{total time}}$ (m/s). Average velocity $= \dfrac{\text{displacement}}{\text{time}}$.
- Displacement $\leq$ distance; for a full round trip displacement and average velocity are zero.
- $1\ \text{km/h} = \dfrac{5}{18}\ \text{m/s}$.

Acceleration

- $a = \dfrac{v - u}{t}$ (SI unit m/s^2); positive when the speed increases, negative (retardation) when it decreases.
- Uniform motion: equal distances in equal times (zero acceleration). Non-uniform: unequal distances in equal times.

Equations of motion (uniform acceleration)

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$
- u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement.

Graphs

- Position-time graph: slope = velocity; straight line = uniform velocity; curve = changing velocity; horizontal line = at rest.
- Velocity-time graph: slope = acceleration; area under the graph = displacement; horizontal line = uniform velocity; straight sloping line = uniform acceleration.
- Distance-time graph: slope = speed (never negative).

Motion under gravity (free fall)

- Acceleration $g \approx 10\ \text{m/s}^2$ (9.8 m/s^2) directed downward for all bodies in the absence of air resistance.
- Thrown upward with speed u: maximum height $H = \dfrac{u^2}{2g}$, time of ascent $= \dfrac{u}{g}$; at the highest point $v = 0$ but $a = g$.
- Dropped from rest: $v = gt$, $s = \frac{1}{2}gt^2$.

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